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Prof. Roland Rad
Institute of Molecular Oncology and Functional Genomics
TUM Klinikum Rechts der Isar
Munich, Germany

Re: PhD / Postdoc Position — Cancer Biology and/or Bioinformatics (26_05_001)

Dear Prof. Rad,

I am applying for a position in the Institute of Molecular Oncology and Functional Genomics, specifically in the area of computational large-scale data integration. My background is computational biology with doctoral research at TUM, and the tools I have built for large-scale genomic data analysis are publicly deployed. The sequence homology search tool built on the Gospel framework is available at vivid-symbolism.vercel.app/search and can be used directly without installation.

Large-scale genomic data integration. Gospel is a whole-genome variant analysis and RNA-seq integration framework designed for the scale that massively parallel CRISPR and transposon screens generate. It processes million-scale VCF files at 10^6 variants per second using Bayesian network inference, Gaussian mixture models, and variational Bayes for multi-omics integration. It has been validated against 1000 Genomes Phase 3, gnomAD v3.1.2, and UK Biobank, and handles the full pipeline from variant calling and RNA-seq quantification through pathway analysis (KEGG, Reactome) and functional annotation. The multi-dimensional integration problem that arises when combining CRISPR screen readouts, somatic mutation profiles, transcriptomics, and clinical outcomes is precisely what the framework was designed for.

Cancer biology and immune mechanisms. My manuscript *On the Thermodynamic Consequences of Categorical Completion on Biological Systems* develops a model of adaptive immune selectivity that predicts cancer-associated proteins as primary targets from structural first principles. MHC binding affinity correlates with the categorical richness of the parent protein's isoform and post-translational modification landscape ($r = 0.73$, $p < 10^{-12}$, $n = 2,847$ IEDB peptides); the model achieves AUC 0.81 against NetMHCpan's 0.68, and explicitly predicts collagen isoforms and tumour-associated antigens as primary immune targets. A companion paper, *On Disease Epistemology in Blind Receiver*, proves that loop-holonomy self-consistency is the unique viable diagnostic criterion for disease state identification (AUC = 1.00, 25/25 validation experiments) and provides a sparse ℓ_1 linear program for therapeutic target selection with minimum off-target effects; it is affiliated with the AImE Registry and TUM School of Life Sciences Weihenstephan.

Cancer cell state transitions. The Syndrome framework models cellular state through

eight oscillator classes (protein, enzyme, membrane, channel, ATP, genetic expression, calcium, circadian), each with a coherence index $\eta_i \in [0, 1]$, computing a disease vector $\mathbf{D} \in [0, 1]^8$ that locates the cell's current state in a partition geometry on the $[0, 1]^8$ hypercube. Trajectory computation runs in $\mathcal{O}(N \log N)$; therapeutic target selection uses sparse LP to identify the minimum-intervention path from disease state to healthy attractor. Applied to cancer biology, the framework addresses cell fate transitions and metastatic progression as trajectory completion problems within this state space — a natural extension of the institute's interest in cell fate transitions and the molecular mechanisms of metastasis.

Multi-omics and imaging. For the metabolomics and imaging dimensions of multi-dimensional data integration, lavoisier.vercel.app/experiment provides a virtual mass spectrometry instrument running force-free simulation in the browser, with a downloadable Rust executable for large-scale local runs. The underlying framework processes mzML datasets exceeding 100 GB via Ray distributed computing and reaches 98.9% feature extraction accuracy against the NIST library.

Institutional context. I completed doctoral research in Computational Lipidomics at TUM / Bayerisches Forschungsinstitut, where I developed mass spectrometry pipelines and federated learning architectures for distributed clinical analysis. Published work is affiliated with the AIME Registry and TUM School of Life Sciences Weihenstephan. I am legally resident in Germany with full work authorisation and am available immediately. English is native; German is conversational.

Yours sincerely,

Kundai Farai Sachikonye